A Free Rescore Is Not Free: A Benchmark-Recommended Score Substitution Reverses Sign Across Medical Imaging Datasets

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Abstract

Reconstruction-based anomaly detectors bundle two separable decisions: the objective the model is trained under, and the rule that turns the residual into a score. The literature moves them together. A current benchmark recommends substituting a structural-similarity residual into pixel-trained models, reporting that it improved every model it was applied to. Crossing four training objectives with five anomaly scores across three datasets (RSNA, VinDr-CXR, LAG; 64×64, three seeds), we measure it against the pixel score on identical weights: +9.08 AUROC on RSNA and −10.54 on LAG, both $p_{\mathrm{Holm}} < 10^{-30}$ under paired DeLong, and a non-significant −2.37 on VinDr-CXR. Turning that factorial on the result itself, the substitution is not one decision: it resolves into a change of distance and a loss of scored support — the pixels the score averages over — with support alone costing about 2 to 4 points on every dataset. At matched support VinDr-CXR is a null. The distance component's sign depends jointly on the dataset and on a Gaussian bandwidth we find no benchmark ablates. The effect is monotone in that bandwidth, and each dataset changes sign at a different one, with the library default inside that spread: below it the substitution hurts on all three, above it helps on all three. On LAG it removes 43% of the cases flagged at 95% specificity. We release per-image scores and trained weights so a new objective or score can be placed in the grid without retraining.

Keywords: anomaly detection, medical imaging, evaluation, reproducibility, autoencoders

Data and Code Availability This study uses three public, de-identified benchmark datasets in the preprocessed form released with the MedIAnomaly benchmark (Cai et al., 2025): the RSNA Pneumonia Detection Challenge (Shih et al., 2019), VinDr-CXR (Nguyen et al., 2022), and LAG (Li et al., 2019). No new data were collected. Anonymized code, per-run manifests, the per-image anomaly score vectors for every reported cell, and the trained weights are provided as supplemental material. The score vectors reproduce every reported comparison and significance test; placing a *new* anomaly score in the grid additionally requires the released weights, since it needs the reconstructions, but no retraining. The variance-head diagnostics require a forward pass through the released head weights.

Institutional Review Board (IRB) This research does not require IRB approval. It reanalyses three public imaging benchmarks, all fully de-identified, all released for methodological research, and none carrying identifiable private information. We recruited no human subjects, accessed no protected health information, and ran no clinical intervention or deployment.

Use of AI Assistance We used a large language model as a coding and writing assistant: to help implement and review the experiment harness, to draft and revise prose, and to cross-check claims against our experimental record. It was not part of the method under study and produced no experimental result. The authors verified every reported number against the stored per-run outputs, read and confirmed every cited work, and take full responsibility for the content.

1. Introduction

Unsupervised anomaly detection promises something a supervised classifier cannot offer in medicine: a detector trained only on normal images, which does not require an annotated example of every pathology it must eventually flag. The dominant recipe is reconstruction — train an autoencoder on healthy images, and treat whatever it fails to reproduce as evidence of abnormality. A decade of work has produced a long list of named methods built on that recipe, and benchmarks that rank them (Baur et al., 2021; Lagogiannis et al., 2024; Cai et al., 2025).

The setting is not incidental. Unsupervised detection is proposed where labels run out: triaging a worklist no reader can clear, or screening for a condition nobody has annotated examples of. More than half of prevalent glaucoma is undetected in every region studied (Soh et al., 2021), and a team building a screen for it has every reason to adopt a change a benchmark reports as free. On LAG — a fundus collection labelled by specialists reading intra-ocular pressure, visual field and optic disc together — one such change removes 43% of the cases the detector had been flagging at a fixed false-positive rate. Sensitivity at 95% specificity falls from 22.5% to 12.8%. Neither figure is deployable and we make no deployment claim; the point is what a scoring rule silently decides.

The ranking hides a design decision. A reconstruction-based detector uses a distance function twice: once as the training objective, where it determines what the model is rewarded for reproducing, and again as the anomaly score, where it determines how the residual becomes a number. Nothing requires the two to be the same function. In published work they almost always are, and they are changed together. Bergmann et al. (2019) state the design plainly — their contribution comes "by adapting the loss *and* evaluation functions" — and then attribute the resulting improvement, from 0.688 to 0.966 AUC, to "merely changing the loss function". Two things moved; one was credited. The practice has not gone away: Fan et al. (2025) substitute SSIM for MSE in both roles at once in 2025.

If the two roles are never separated, no reported gain in this family can be assigned to either. Cai et al. (2025) come closest, tabulating the training loss and the anomaly score as separate columns, but every reconstruction method they evaluate instantiates the two with the same distance. The off-diagonal has been entered once, and not as an experiment: Lagogiannis et al. (2024) substitute an SSIM residual into four models trained under pixel-space objectives and report that this "increased the performance of all selected models it was applied to". It is a sensible design decision, honestly described, and offered to readers as a default.

We test it. Reproducing ten methods under one protocol, we cross four training objectives with five anomaly scores on one backbone, so that either decision can move while the other stays fixed. Rescoring costs no training: only the function applied to a fixed residual changes. On three datasets — two chest radiograph collections and one retinal fundus collection — the substitution improves AUROC by 9.08 points on RSNA and harms it by 10.54 on LAG, both Holm-significant, with VinDr-CXR negative on every seed but not separable from noise.

Nor is it one decision. It changes the distance, the bandwidth at which that distance is evaluated, and — because the standard implementation uses valid convolution — the region scored. Unbundled on identical weights, the reported figures resolve into a distance change and a support loss, and VinDr-CXR's negative sign turns out to be the support change. The distance component's sign depends on the dataset and on a Gaussian bandwidth we find no benchmark ablates: each dataset crosses zero at a different bandwidth, the library default falls inside that spread, and outside it the substitution moves the same way everywhere. A free rescore is not free, and it is not one choice.

That generalises into the paper's organising question: which conclusions about these design decisions survive a change of dataset? Applying seven design-level comparisons unchanged to all three, three replicate, three reverse, and one holds its sign everywhere without reaching significance on RSNA. Among the reversals is a mechanism we proposed ourselves — that the advantage of jointly trained uncertainty estimation over a post-hoc head reflects co-adaptation during training — which predicts a gap on every dataset, and LAG has none. Alongside what fails to transfer, a configuration chosen on RSNA and applied unchanged to the two datasets it was not chosen on gains 1.33 AUROC over the strongest single method (seed-level 95% CI [0.41, 2.25]).

Contributions.

- We correct a published recommendation. A benchmark-endorsed score substitution that needs no retraining reverses sign across datasets, significantly in both directions, with the third a null. Unbundled, it is a distance change plus a loss of scored support, and the two sum to the reported figures exactly (Section 5).
- A replication table for design decisions across three datasets: three comparisons replicate, three reverse, and one holds its sign everywhere without reaching significance on RSNA. We report all three outcomes with equal prominence.
- The objective-by-score factorial, used as an attribution instrument. The two decisions do not

add: changing both exceeds the sum of the single changes by 9.8 AUROC points. Mismatching them does active harm — a perceptually trained model read in pixel space scores below chance.

- A measured mechanism for one axis: under an SSIM objective the inherited decoder drifts outside the input range on all three datasets, with a confirmed spatial signature.
- A frozen-backbone variance-head probe, a held-out transfer result, and a released instrument for placing a new objective or score in the grid.

Scope. Every comparison here is internal to image-reconstruction detectors at 64×64 with image-level labels, on one inherited backbone. The best accuracy on this RSNA split belongs to a feature-reconstruction method using ImageNet weights (Meissen et al., 2023), and we do not reach it. That method changes the representation and the score in one step — precisely the design our results show cannot be attributed.

2. Related Work

We group prior work by which design decision it varied, since that is the axis this paper adds to.

Varying the architecture, holding the residual fixed. The comparative studies that established this field vary what the model is, not how its output is read. Baur et al. (2021) evaluate a dozen autoencoder, VAE and GAN variants on four brain MRI datasets under a unified pipeline, with an ℓ_1 residual throughout. They do vary the extraction procedure — Monte-Carlo consensus, gradients, iterative restoration — but that changes how the residual is *obtained*, not the distance that measures it. Lagogiannis et al. (2024) extend the audit across modalities and resolutions, arguing preprocessing and resolution dominate method identity.

Varying the score, holding the objective fixed. That a trained model admits more than one readout is not new, and we do not claim it. Zimmerer et al. (2019) keep one variational autoencoder and derive several scores from different terms of its evidence lower bound. That is a genuine decoupling, and it establishes the score as a free variable — but along a different axis from ours: which part of one objective to read, not which distance does the reading. Meissen et al. (2022) argue from the other direction that the pixel residual is a poor score in the first place, being dominated by error on normal structure rather than pathology. That motivates our question without answering it: it shows the residual is problematic, not which of its two roles is responsible.

Varying the distance function — on both axes at once. Prior work does vary the reconstruction metric, but almost always in both roles at once. Bergmann et al. (2019) adapt "the loss *and* evaluation functions" and then credit the loss alone; Fan et al. (2025) substitute SSIM for mean squared error in both roles in a 2025 video study. Cai et al. (2025) come closest, tabulating the training loss and the anomaly score as distinct columns and noting that earlier studies "do not explicitly compare the effects of different distance functions". But every reconstruction method they evaluate holds the same distance in both columns, so their grid is populated only on its diagonal. For the objective alone they do report that a perceptual loss beats ℓ_2 on several datasets and loses on another — dataset-dependence, established for the training role.

The one off-diagonal cell in the literature. Lagogiannis et al. (2024) enter the off-diagonal directly, replacing the pixel-wise absolute error with a structural similarity residual at scoring time for models that optimise pixel-space objectives, on the grounds that SSIM accounts for texture and structure and not only luminance. To our knowledge it is the only published instance in this literature of training under one distance and scoring under another. It is reported as a preprocessing decision rather than as a result, and it carries a universal claim.

What remains open. No study we could find, in medical, industrial or video anomaly detection, crosses the objective with the score. Behrouzi et al. (2023) come close: industrial thermal autoencoders trained under both MSE and SSIM objectives, then read under three anomaly scores. But they index their results by score alone, so no loss-by-score attribution can be recovered. Without the cross, a reported gain in this family belongs to the objective, to the score or to their interaction, and nothing in the report says which — nor, as we show, whether it keeps its sign on the next dataset.

3. Method

Two roles for one distance. A reconstruction-based detector is a map f fitted on normal images and a scoring functional applied to its output. Fitting minimises a distance $d_{\text{train}}(x, f(x))$ over normal data; scoring assigns $s(x) = d_{\text{score}}(x, f(x))$, larger meaning more anomalous. Published reconstruction-distance methods set $d_{\text{train}} = d_{\text{score}}$ and report one number; the gradient scores of Zimmerer et al. (2019) are the exception, and they vary which term of one objective is read rather than which distance reads it. We tie neither.

The grid. We cross four objectives — ℓ_2, SSIM, a relative perceptual distance on VGG-19 features up to `relu4_2`, and a heteroscedastic Gaussian likelihood — with five scores: the same four, plus the gradient-based score of Zimmerer et al. (2019). SSIM uses the reference implementation's defaults throughout: an 11×11 Gaussian of σ=1.5, C_1=0.01^2 and C_2=0.03^2 on inputs rescaled to $[0, 1]$, evaluated under valid convolution. Both conventions are inherited without examination, and Section 5 shows each carries part of the effect otherwise attributed to the distance. Figure 1 gives the design and names the published method each diagonal cell instantiates. Of $4 \times 5 = 20$ cells, 17 are defined. The three that are not are the uncertainty score applied to the three objectives that emit no variance; Section 3 supplies a proxy by fitting a variance head to the frozen backbone, which consumes training data and is therefore not a rescore.

Off-diagonal cells need no training. We load the weights frozen and change only the scoring functional, so a row costs one training run and four evaluations, and every such cell is reproducible from the released weights.

A variance head on a frozen backbone. The uncertainty score $s(x) = \text{mean}[\exp(-\log \sigma^2)\,(x - \hat{x})^2]$ requires a per-pixel variance that only a heteroscedastic model emits. To obtain it for backbones that do not, we attach a small convolutional head to an *already-trained, fully frozen* autoencoder and fit only the head, on normal training images, under the heteroscedastic Gaussian negative log-likelihood (Nix and Weigend, 1994; Kendall and Gal, 2017). Every weight is frozen and the module re-asserted into evaluation mode each epoch, so the reconstruction is bit-identical to the parent run. At w=2 the head is 14,528 parameters, 0.62% of the backbone.

Three controls follow. A capacity sweep (Section 4.1), where w=1 — a single transposed convolution with normalisation and nonlinearity stripped — fails everywhere, making the threshold one of nonlinearity rather than capacity. A backbone-transfer control fits the same head to a perceptually trained backbone, where it fails by 30.6 AUROC, bounding the claim to pixel-space objectives. And the head is compared both against its parent and against a jointly trained model, so every result is labelled with its parent backbone.

4. Experimental Setup

Inherited protocol. Splits, backbone, preprocessing and every hyperparameter come unchanged from Cai et al. (2025): 64×64 inputs scaled to $[-1, 1]$, a 16-dimensional latent, 250 epochs, Adam at 10^{-3}, batch size 64, three seeds. We inherit rather than tune: it is what makes the reproduction valid and the factorial confounder-free, since within the grid only the objective and the score differ.

Datasets. RSNA (Shih et al., 2019) and VinDr-CXR (Nguyen et al., 2022) are chest radiograph collections; LAG (Li et al., 2019) is retinal fundus photography with glaucoma labels assigned by specialists from intra-ocular pressure, visual field and optic disc assessment. Training uses normal images only: 3851, 4000 and 1500 images respectively. Test sets are balanced by construction — 1000/1000 for RSNA and VinDr-CXR, 811/811 for LAG — a benchmark artifact rather than a clinical one, to which we return in Section 5. Splits are the benchmark's own, used unchanged: no image occurs in both halves and no preprocessing statistic is fitted on test data. The published split file does not record patient identity, so patient-level separation cannot be verified from it.

Reproduction as calibration. Before crossing anything we reproduced ten published methods under this protocol (Section C): plain, ℓ_1, SSIM (Bergmann et al., 2019), perceptual (Shvetsova et al., 2021), variational (Kingma and Welling, 2014), uncertainty (Mao et al., 2020) and denoising (Kascenas et al., 2023) autoencoders, with the gradient scores of Zimmerer et al. (2019). Nine of ten fall within the pooled standard deviation of our runs and the published values, the band appropriate to a difference that varies on both sides, and four are within 0.2 AUROC. The exception is a denoising autoencoder run at a single seed, which we report rather than drop. Per-method values and the tighter within-published and within-ours counts are in Section C. Reproducing is not itself a contribution. It is the calibration that licenses everything after it: a grid cell compares to a published number only if the diagonal reproduces it.

Held-out configuration. Two choices are ours rather than inherited: the head width and the ensemble membership. Both were fixed on RSNA, written to a machine-readable selection card, and applied unchanged to VinDr-CXR and LAG, which were run afterwards; no entry was revised. Everything not on that card is inherited, so the selection surface is one width and one membership set. The card and the per-seed held-out records written at run time are released with the code.

Statistics. We report the unit of analysis for every test. Paired DeLong tests (DeLong et al., 1988; Sun and Xu, 2014) treat the test *images* as the sampling unit; our three seeds are scored on one test set, so the mean difference over seeds uses a single covariance over all six score vectors with the contrast $(\mathbf{1}, -\mathbf{1})/3$. Pooling seed by seed instead would assume an independence the design does not have. DeLong carries no seed-variance component, so it says nothing about retraining. Seed-level paired t-tests do, but at two degrees of freedom the 95% multiplier is 4.30 against DeLong's 1.96. They are therefore underpowered for the one- and two-point differences this paper reports, and a null there means unresolved rather than equivalent. Holm correction (Holm, 1979) is applied within each dataset, each being a separate replication; the family is the seven comparisons of Table 1. Sensitivity at fixed specificity uses a split-half protocol — threshold on one half of the test set, sensitivity on the other, over 500 stratified splits — with paired image-level bootstrap intervals recomputing the threshold inside each replicate. We construct no t-intervals over three seeds; per-seed values are reported instead.

4.1. Selection discipline

One quantity here was fixed by looking at test performance: the variance head's width, set to w=2 after inspecting RSNA test AUROC. A paper arguing that design decisions are validated on the wrong data should state its own instance. The sweep (Section D) shows the inspected quantity is flat, that w=4 beats the inherited w=2 on both held-out datasets so the choice cost rather than flattered them, and that a label-free rule selects w=8, which carried unchanged averages 0.57 AUROC above w=2. The width does not enter the transfer result.

5. Results

5.1. The substitution reverses, and what that costs

Rescoring a trained autoencoder with SSIM changes nothing about the model: the weights are the parent run's, the reconstruction is bit-identical, and only the function applied to the residual differs. That change is worth +9.08 AUROC on RSNA and −10.54 on LAG, both Holm-significant at $p < 10^{-30}$, and −2.37 on VinDr-CXR, negative on every seed but not separable from image-sampling noise ($p_{\text{Holm}} = 0.068$; Table 1). A substitution reported to improve every model it was applied to improves one of our three datasets, does nothing measurable on a second, and significantly harms a third.

It is not seed noise: within each dataset all three seeds agree in sign, and the RSNA and LAG effects dwarf the seed-level spread. It is also not one decision, and the instrument this paper is built on says so. Replacing a pixel residual with an SSIM residual changes the distance, the bandwidth at which that distance is evaluated, and — because the standard implementation evaluates SSIM under valid convolution — the region scored. The last matters because the discarded border is not empty: restricting the pixel score to the same central region costs 4.84, 3.24 and 3.52 AUROC. We separate the two by rescoring through an intermediate. Writing $\mathcal{A}(s)$ for the AUROC of score s, with s_{ℓ_2} the pixel score, s_{val} the SSIM score as implemented and s_{pad} the same score replicate-padded to full support, $\mathcal{A}(s_{\text{val}}) - \mathcal{A}(s_{\ell_2}) = [\mathcal{A}(s_{\text{pad}}) - \mathcal{A}(s_{\ell_2})] + [\mathcal{A}(s_{\text{val}}) - \mathcal{A}(s_{\text{pad}})]$. The split is an identity, true of any intermediate; what earns the two terms their names is that s_{pad} changes the distance at fixed support. They are +11.13, −0.19 and −6.50, and −2.06, −2.19 and −4.04, re-adding to the headline up to two-decimal rounding. Components shift by up to 1.3 points under reflect or zero padding, so we name the convention wherever we quote them; every convention gives the same total. The RSNA–LAG reversal belongs to the distance and survives. VinDr-CXR's does not: at matched support it is a null, and its reported figure is almost entirely support. We found this by turning our own instrument on our own headline.

It is also not an artefact of the summary metric. Estimating the threshold at 95% specificity on one half of the test set and measuring sensitivity on the other, over 500 stratified splits, the substitution moves sensitivity from 17.9% to 31.5% on RSNA, 13.9% to 11.0% on VinDr-CXR and 22.5% to 12.8% on LAG. Neither LAG figure is deployable — published glaucoma screening operates near 85% sensitivity here — and we make no deployment claim.

Nor is the reversal an artefact of the SSIM window's size relative to a 64×64 image. Recomputing the score on identical weights at smaller windows deepens the harm on LAG while eroding the gain on RSNA, and the sign disagreement survives to 5×5; varying the Gaussian bandwidth itself, the sign depends on it as much as on the dataset (Section 7).

5.2. What replicates and what reverses

Applying seven design-level comparisons unchanged to all three datasets, three replicate in sign with Holm significance everywhere, three reverse, and one holds its sign on all three without clearing Holm on RSNA (Table 1). We call a comparison replicated when it agrees in sign *and* clears Holm on all three datasets. The third category is not a technicality: the head improves on frozen AE-SSIM by +15.92 on LAG and +13.95 on VinDr-CXR but only +1.19 on RSNA, where it misses significance. Sign is more portable than magnitude, and magnitude than significance.

Grouping the rows by the decision under test shows the instability is not spread evenly: every comparison holding its sign on all three measures a construction against something it is built on top of, and every reversal compares arms where neither contains the other. We return to this in Section 6.

Table 1: Design-level comparisons applied unchanged to three datasets. Values are AUROC point differences; positive favours the first-named method; per-seed values are in Section A. Paired DeLong on the test images with the multi-seed contrast of Section 4, Holm-corrected within each dataset. * $p_{\text{Holm}} < 0.05$; an unmarked cell is not evidence of absence, and all three unmarked here are sign-consistent across seeds. DeLong carries no seed-variance component; per-seed values are in the appendix. Three comparisons replicate with significance on all three datasets, three reverse, and one is sign-consistent without clearing Holm on RSNA.

Comparison	RSNA	VinDr	LAG
Score substitution			
AE rescored w/ SSIM vs. AE	$+9.08^{*}$	-2.37	-10.54^{*}
Head vs. the backbone it is fitted onto			
head vs. frozen AE	$+13.65^{*}$	$+12.38^{*}$	$+4.31^{*}$
head vs. frozen AE-SSIM	$+1.19$	$+13.95^{*}$	$+15.92^{*}$
Head vs. joint training			
AE-U vs. head on AE	$+4.86^{*}$	$+5.72^{*}$	-0.78
head on AE-SSIM vs. AE-U	$+0.13$	-1.03	-4.35^{*}
Score ensembling			
ensemble vs. AE-PL	$+1.16^{*}$	$+1.57^{*}$	$+1.08^{*}$
ensemble vs. AE-U	$+1.88^{*}$	$+2.95^{*}$	$+4.49^{*}$

5.3. Attribution: what the objective buys

Figure 1 gives the full grid on RSNA. The four boxed cells are the diagonal: methods whose anomaly score is the same *distance between x and $\hat{x}$* that trained them. Every reconstruction-distance method in this family sits there. The gradient scores are the published exception, and they are the decoupling of Zimmerer et al. (2019): a different term of one fixed objective, not a different distance.

The two decisions are not additive. Against the plain autoencoder, changing only the score to SSIM is worth +9.08, changing only the objective −6.45, and doing both — which is AE-SSIM — +12.46: the joint effect exceeds the sum of the single changes by 9.8 points. We report this as a descriptive additivity check in AUROC points relative to a fixed reference, not as an analysis of variance: it carries no error model, and AUROC is bounded and not additive. The same arithmetic for the perceptual pair gives 35.2 points, but that figure is inflated by a constituent cell sitting below chance, where AUROC's geometry differs, and we do not lean on it.

The perceptual row shows what this costs: read with ℓ_2 it gives 44.9 and with SSIM 44.0, both below chance, and both replicate (38.7 and 34.7 on VinDr-CXR). The reading is asymmetric. A score can be informative in directions the objective never constrained — on RSNA the ℓ_2-trained model is read best by SSIM — yet an objective can leave a direction so unconstrained that reading it inverts the ranking. Mismatching is not uniformly costly but unpredictable in direction and magnitude, which is what makes an inherited scoring rule unsafe. Below chance is not the absence of signal but its inversion: 44.9 carries as much information as 55.1, with the sign reversed. A scoring rule inherited without val-

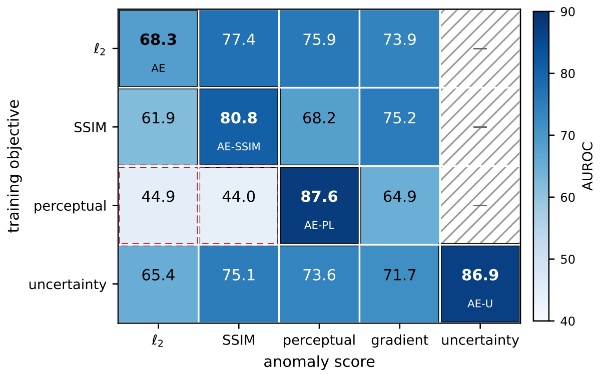

Figure 1: The objective × score grid on RSNA, AUROC averaged over three seeds. Boxed cells are the diagonal and name the published method they instantiate. Red dashed cells are below chance. Hatched cells are undefined: the uncertainty score requires a per-pixel variance that only the heteroscedastic objective emits, and Section 3 supplies a proxy by fitting a variance head to a frozen backbone — not a rescore, since it consumes training data. The grid was run on all three datasets at three seeds; RSNA is shown here and the other two are in Section B.

idation may not merely underperform; it may rank the wrong images first, and nothing in the score itself reveals which.

5.4. What the uncertainty score computes

One column of the grid cannot be filled by rescoring: the uncertainty score needs a per-pixel variance that only its own objective emits. A variance head fitted to the frozen backbone supplies a proxy, reaching 82.00 on RSNA against the plain autoencoder's 68.35, but it consumes training data and reads internal decoder features, so it is not a score substitution. Four controls in Section F show what it learns is a per-image calibration rather than a spatial uncertainty map; two readings of the score are consistent with our measurements and we have not separated them.

5.5. Held-out transfer, and its failed extension

If design decisions mostly fail to carry across datasets, the question is whether anything does. Answering it means fixing a configuration before seeing the datasets that will judge it. Two choices were fixed on RSNA and applied unchanged elsewhere. A rank-average of the perceptual and uncertainty scores gains 1.33 AUROC over the strongest single method on the two datasets it was not fixed on: six of six seed measurements positive, seed-level 95% CI $[0.41, 2.25]$. With dataset as the unit of analysis $n = 2$, which supports the agreement in sign and no p-value.

The same protocol refuses an extension: a third member gains +0.72 on RSNA, where it was selected, and loses 1.26 on held-out LAG. A gain measured where a choice was made is not evidence that the choice transfers.

6. Discussion

SSIM training leaves the input range. One mechanism here is measured rather than inferred. Inputs lie in $[-1, 1]$, so an in-range reconstruction cannot give a squared residual above 4; under an SSIM objective the largest is 14.6, 12.1 and 274.6, with LAG reconstructions reaching -17.2. Two known properties of SSIM (Wang et al., 2004; Pambrun and Noumeir, 2015; Nilsson and Åkenine-Möller, 2020) produce it. The structure term is offset-invariant and the luminance restoring force scales with local mean intensity, so dark regions are barely anchored. The implementation also clamps before evaluating, so the gradient outside the range is exactly zero and nothing pulls back a pixel that has left. Drift should therefore be downward and confined to dark regions, and it is: on every dataset and every seed, between 95.7% and 99.6% of out-of-range pixels lie in the darkest quintile of local input intensity, and reconstructions run below the range, never above. An ℓ_2 objective stays in range everywhere (1.7, 1.6, 0.7; Section E), which is why a pixel score on an SSIM-trained model reads the drift rather than the pathology. That the same code reproduces AE-SSIM's published RSNA AUROC to within 0.1 points while exhibiting the violation argues against an implementation fault. The properties are known; what appears unreported is that they make the divergence invisible to every number the benchmark records, loss and score alike being clamped.

A parallel we decline to explain. SSIM enters this study twice and both entries move together: as a score it is worth +9.08, -2.37 and -10.54 against the plain autoencoder, and as an objective +12.46, -1.56 and -11.61 — agreeing in sign on all three and within 3.4 points. We decline to call these one phenomenon. The range mechanism belongs to the objective axis alone, since the score-axis comparison runs on the in-range ℓ_2 backbone. Nor does range explain the pattern even there: RSNA and VinDr-CXR diverge by similar amounts (14.6 and 12.1), yet the objective is worth +12.46 on the first and −1.56 on the second, and the second is not separable from noise ($p = 0.16$). A numerical parallel across three datasets is the evidence this paper argues is insufficient for attribution.

Where the instability lives. The partition in Table 1 aligns exactly with whether one arm is built on top of the other: all four comparisons that hold their sign on every dataset are additive, and all three that reverse compare arms where neither contains the other. We noticed this after the fact, on seven comparisons chosen for other reasons, so we state it as a checkable hypothesis rather than a result (Section A).

What follows for practice. A scoring rule is not a property of a method that travels with it, and a rescore reported as free is a bundle of decisions. Revalidate it on labelled data from the target distribution; where none exists, do not adopt it. The converse is worth as much: additive constructions held their sign everywhere we tried, and are the one class of change this evidence supports carrying unvalidated. We do not explain why the distance's sign depends on the dataset.

7. Limitations

1. Resolution, and a windowed score. All comparisons are at 64×64, inherited from the benchmark, which reports this "achieves comparable performance to using a larger input size" (Cai et al., 2025) — but that ablation varies resolution under pixel objectives and pixel scores, and we find no published ablation covering a *windowed* score. Tested at fixed resolution (Section D), the effect is monotone in the SSIM bandwidth on all three datasets, losing everywhere at σ=0.5 and helping everywhere by σ=3. Each dataset changes sign at a different bandwidth — between 0.85 and 1.0 on RSNA, 1.5 and 2.0 on VinDr-CXR, 2.0 and 3.0 on LAG — and the library default falls inside that spread. The 128 × 128 equivalent of that default, σ=0.75, is negative on all three (−3.31, −8.77, −18.39), so we predict the RSNA gain vanishes rather than merely shrinks. Resolution also changes image content, so the replication remains the first item of further work.

2. Three datasets, and modality is confounded with the sign. We claim no general law of transfer, only one specific failure of it: a significant gain on one dataset and a significant loss on another. But once VinDr-CXR is read as a null at matched support, the panel is one positive and one null chest radiograph collection and one negative fundus collection — equally consistent with the distance simply suiting radiographs and not fundus images. A null is weak evidence against a strong modality effect and none against a weak one; separating them requires a dataset that is neither, and the panel does not contain one.

3. Three seeds. DeLong tests condition on these trained models and carry no seed-variance component, so claims below roughly two AUROC points rest on the seed-level column, where three seeds are underpowered. A non-significant seed-level result is unresolved, not null: the LAG comparison of AE-U against the head (−0.78, $p_{\text{Holm}} = 0.13$) is reported as direction unresolved.

4. One backbone, image-level labels. Inheriting the architecture and hyperparameters is what makes the reproduction valid and the factorial free of confounders. But we have not tested whether the decomposition holds under a different backbone, and we make no localisation claim.

5. Balanced test sets. AUROC is prevalence-invariant, so the headline is unaffected. But we read the operating points on a 50% test set (Section G), and those sensitivities do not transport to a screening cohort. LAG is a tertiary referral collection, whose case mix is more severe than a screening population's.

6. Selection, and one incomplete cell. One quantity was fixed by inspecting test performance (Section 4.1). On LAG the head fitted to a frozen AE-SSIM backbone overflowed on one seed, so that comparison uses two seeds, marked wherever it appears. Since all three LAG AE-SSIM backbones leave the input range, every AE-SSIM-on-LAG number is measured on a diverged reconstruction — unevenly, since the SSIM score is evaluated on the clamped reconstruction and cannot see the drift while the pixel score reads it directly.

8. Conclusion

Two decisions inside a reconstruction-based detector have been moving together, and the literature has been crediting one of them. Separating them shows that a substitution a current benchmark recommends as free gains 9.08 AUROC points on one dataset and loses 10.54 on another — and that unbundled, into a distance, a bandwidth and a region scored, it was never one decision either. We offer not a better detector but the instrument that says which part of one to believe.

Appendix A. Per-seed differences

Every comparison in Table 1, broken out into its three individual seed measurements: AUROC points, paired DeLong on identical images. The main text reports the means; these are the values behind them. Two exceptions the means conceal are worth naming. One RSNA seed for the head against frozen AE-SSIM is negative. And the two comparisons of a post-hoc construction against a jointly trained model disagree in sign *within* a dataset, which is why the LAG row of the former reads direction unresolved rather than no difference. Across the four comparisons that hold their sign on all three datasets, 35 of 36 seed measurements agree in sign.

The additive/substitutive partition. Section 6 names this partition: every comparison that holds its sign is additive, every reversal compares arms where neither contains the other. Two details belong here rather than there. First, the counts differ because the criteria do — one additive comparison, the head against a frozen AE-SSIM, holds its sign on all three datasets but misses Holm on RSNA, so four hold their sign where three replicate. Second, the hypothesis is checkable, which is why we state it: it predicts that a further additive comparison should hold its sign on a fourth dataset, and that a further substitution should not. The co-adaptation account we proposed for the joint-versus-post-hoc gap is a casualty of the same partition — it predicts a gap on every dataset, and LAG has none.

Table 2: Per-seed AUROC differences (seeds 42, 43, 44). A dash marks the single post-hoc head run on LAG that failed a non-finite-loss guard and was not recorded.

	seed 42	seed 43	seed 44
AE rescored with SSIM vs. AE			
RSNA	+8.80	+8.96	+9.47
VinCXR	−1.86	−2.77	−2.50
LAG	−9.55	−10.93	−11.13
head vs. frozen plain AE			
RSNA	+13.87	+13.75	+13.33
VinCXR	+13.31	+12.50	+11.34
LAG	+1.94	+5.47	+5.52
head vs. frozen AE-SSIM			
RSNA	+1.29	+2.58	−0.30
VinCXR	+14.88	+14.84	+12.12
LAG	+13.49	+16.93	+17.35
AE-U vs. head on frozen AE			
RSNA	+4.16	+3.98	+6.46
VinCXR	+5.45	+6.98	+4.74
LAG	+0.48	−2.50	−0.32
head on AE-SSIM vs. AE-U			
RSNA	+1.47	−0.16	−0.92
VinCXR	−1.32	−3.24	+1.48
LAG	−0.13	—	−8.57
ensemble vs. AE-PL			
RSNA	+0.90	+1.10	+1.49
VinCXR	+1.96	+2.52	+0.25
LAG	+0.71	+0.77	+1.76
ensemble vs. AE-U			
RSNA	+2.34	+1.97	+1.32
VinCXR	+2.32	+1.90	+4.61
LAG	+6.08	+5.21	+2.17

Appendix B. The factorial on the held-out datasets

Figure 1 shows RSNA. The same 17 cells were run at three seeds on both held-out datasets; they are given here. Rows are training objectives, columns anomaly scores, bold marks the diagonal, and a dash marks the uncertainty score where no variance output exists.

Table 3: VinDr-CXR (upper) and LAG (lower), AUROC averaged over three seeds.

objective	ℓ_2	SSIM	perc.	grad.	unc.
VinDr-CXR					
ℓ_2	**56.03**	53.65	57.80	60.93	—
SSIM	52.51	**54.46**	54.03	58.33	—
perceptual	38.67	34.69	**75.50**	45.81	—
uncertainty	55.71	51.12	53.96	61.26	**74.13**
LAG					
ℓ_2	**78.96**	68.43	66.74	78.67	—
SSIM	46.28	**67.35**	49.77	43.13	—
perceptual	53.90	43.38	**85.90**	54.28	—
uncertainty	77.71	66.65	68.75	82.91	**82.50**

Appendix C. Reproduction of the inherited baselines

Ten published methods under the protocol of Cai et al. (2025), RSNA, image-level AUROC. The reference column is their Table 6 and is used only as a target to check our implementations against. Nine of ten fall within one standard deviation; four within 0.2 points. The exception is the denoising autoencoder, which we ran at a single seed: one seed cannot separate an implementation gap from seed variation, and we report it rather than drop it.

Table 4: Reproduction. Ours is mean ± population standard deviation over seeds; on three seeds this understates the sample standard deviation by a factor of 1.22.

Method	n	ours	reference	Δ
AE	3	68.35 ± 0.65	67.5 ± 0.9	+0.85
AE-ℓ_1	3	68.53 ± 0.86	68.1 ± 0.4	+0.43
AE-SSIM	3	80.81 ± 0.35	80.9 ± 0.3	−0.09
AE-PL	3	87.58 ± 0.27	87.5 ± 0.2	+0.08
VAE	3	67.33 ± 1.41	67.9 ± 0.8	−0.57
AE-U	3	86.86 ± 0.44	86.5 ± 0.9	+0.36
DAE	1	83.87	86.1 ± 0.7	−2.23
AE-Grad	3	73.86 ± 0.87	73.7 ± 1.0	+0.16
VAE-Grad$_{\text{rec}}$	3	71.68 ± 1.11	71.6 ± 0.8	+0.08
VAE-Grad$_{\text{combi}}$	3	67.84 ± 1.35	67.4 ± 0.3	+0.44

Appendix D. Head width and the SSIM window

Width sweep. The post-hoc variance head across five widths, three seeds, all three datasets. Widths 2, 4 and 8 span 0.23, 1.77 and 0.78 AUROC on RSNA, VinDr-CXR and LAG respectively. Width 1 reduces to a single transposed convolution with normalisation and nonlinearity stripped and fails everywhere; width 16 is an optimisation failure, with initial training losses across seeds spanning three orders of magnitude.

Table 5: Post-hoc head AUROC by width. The width used throughout the paper is w=2; see Section 4.1.

	w=1	w=2	w=4	w=8	w=16
RSNA	66.94	82.00	82.05	81.81	78.22
VinDr-CXR	52.49	68.41	70.18	70.03	64.50
LAG	74.04	83.27	83.56	82.78	81.23

Bandwidth and window. Varying the window at fixed σ changes the truncation, not the scale: a σ=1.5 kernel truncated at ±3 retains about 95% of its mass. Shrinking it deepens the harm on LAG while eroding the gain on RSNA, and the sign disagreement survives to 5×5. Varying the bandwidth itself is the operative manipulation. Values are AUROC differences against the pixel score on the same weights; the bandwidth rows use replicate-padded SSIM at full support.

Table 6: Effect of the SSIM rescore by window size at fixed σ=1.5 (upper) and by Gaussian bandwidth (lower). σ=0.75 is the 128×128 scale-equivalent of the library default.

	RSNA	VinDr-CXR	LAG
window, σ fixed at 1.5			
11×11	+9.08	−2.37	−10.54
7×7	+7.99	−3.12	−10.48
5×5	+4.07	−5.20	−13.11
3×3	−5.08	−9.88	−19.49
bandwidth σ, full support			
0.50	−10.03	−11.90	−21.72
0.75	−3.31	−8.77	−18.39
1.00	+2.72	−5.55	−14.07
1.50	+11.13	−0.19	−6.50
2.00	+15.20	+3.28	−1.52
3.00	+18.13	+7.17	+3.55

Appendix E. Reconstruction range by objective

Largest squared residual on 256 normal training images, averaged over three seeds. Inputs lie in $[-1, 1]$, so an in-range reconstruction cannot exceed 4; values above it measure how far the decoder has left the input range. The reference decoder we inherit ends in a transposed convolution with no output activation, so nothing bounds it but the objective.

Table 7: Maximum squared residual by training objective. The ℓ_2 row is the control: its reconstructions stay in range everywhere, which is what makes the score-axis comparisons in Table 1 clean.

objective	RSNA	VinDr-CXR	LAG
ℓ_2 (plain AE)	1.7	1.6	0.7
SSIM	14.6	12.1	274.6
perceptual	3.8	2.8	2.3
in-range maximum		4.0	

The perceptual row is worth noting: AE-PL stays in range on all three datasets, so its below-chance pixel-space cell is not a gross intensity violation. That cell is consistent with contrast compression rather than a global offset, which we do not test further here.

Appendix F. The variance head

What it learns is not error magnitude. On RSNA the full head reaches 82.00 against the plain autoencoder's 68.35; reduced to one scalar per image, discarding all spatial structure, it still reaches 79.87 — so the benefit is per-image calibration, not a spatial uncertainty map. That scalar cannot be a monotone function of the image's own mean residual, or the score would collapse back to 68.35. A static map fixed to the training-set mean residual gives 69.10 and zero-parameter whitening 69.59, so the head is neither an anatomical prior nor a trivial normalisation. Its log-variance rises with an image's own residual ($\rho = 0.51$) while abnormal images receive a *smaller* log-variance than normal ones, weighting them 10.5% higher. One reading is a residual measured relative to expected difficulty; another — a normaliser fitted on normal features, extrapolating downward off that manifold — predicts the same signs, and we have not separated them.

Appendix G. Operating-point protocol

We report sensitivity at fixed specificity under a split-half protocol. We split the test set in half, stratified by label, fix the threshold achieving 95% specificity on the first half, and measure sensitivity for both scoring rules on the second, repeating over 500 stratified splits and averaging per seed. The threshold is therefore never estimated on the data it is evaluated on. Confidence intervals come from a paired image-level bootstrap that recomputes the threshold inside every replicate. Holding it at its point estimate would drop threshold-estimation uncertainty, which dominates when the quantile sits far into the tail. We construct no intervals over three seeds, and report per-seed values instead.

These sensitivities do not transport to a screening cohort, because the test sets are balanced. LAG in particular is drawn from a tertiary referral population whose case mix is more severe than a screening population's, and its labels come from specialists reading intra-ocular pressure, visual field and optic disc together.

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